

Effect of Contact Points Feedback on Two-Thumb Touch Typing in Virtual Reality

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ABSTRACT

Two-thumb touch typing (4T) is a touchpad-based text-entry technique also used in virtual reality (VR) systems. However, the performance of 4T in VR is far below that of 4T in a real environment, such as on a smartphone. Unlike “real 4T”, 4T in VR provides virtual cursors representing the thumb positions determined by a position tracker. The virtual cursor positions may differ from the thumb contact points on an input surface. Still, users may regard them as their thumb contact points. In this case, the virtual cursor movements may conflict with the thumb movements perceived by their proprioception and may contribute to typing errors. We hypothesized that virtual cursors accurately representing the contact points of the thumb can improve the performance of 4T in VR. We designed a method to provide accurate contact point feedback, and showed that accurate contact point feedback has a statistically significant positive effect on the speed of 4T in VR.

CCS CONCEPTS

• Human-centered computing → Text input.

KEYWORDS

virtual reality, text entry, two-thumb touch typing, 4T

ACM Reference Format:

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1 INTRODUCTION

Two-thumb touch typing (4T) is a common text entry technique in virtual reality (VR) [4, 5, 15, 18]. Some previous studies [4, 5, 18] have considered 4T in investigating efficient and preferable text entry methods in VR. VR systems on the market, such as HTC Vive, also provide a 4T interface for handheld controllers. The 4T is an established text entry method in the real world; using two thumbs

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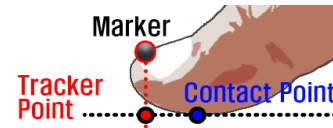


Figure 1: If the position of the virtual cursor is determined by the position of the marker, it may not be the same as the thumb's contact point.

on a smartphone is a representative example. As most people are experts in “real 4T”, it is expected that they may be able to transfer their skill in real 4T to 4T in VR and achieve a high typing speed with 4T in VR. However, the performance of the 4T in VR was significantly lower than that of the real 4T. For example, studies on 4T have shown that real 4T could achieve a typing speed of over 50 words per minute (WPM) [1], but 4T in VR could not achieve 15 WPM [4, 5, 18].

Among the many possible reasons for the performance gap between 4T in VR and real 4T, we focused on the inaccurate visual feedback of thumb positions in VR. In VR, users cannot see the real thumbs but rely on virtual cursors to understand the thumb positions determined by a position tracker (tracker points). In this case, the virtual cursor positions may not be the same as the thumb contact points perceived by users (contact points). The discrepancy between the tracker and contact points is illustrated in Figure 1. We speculate that the virtual cursor movements may conflict with the thumb movements perceived by proprioception and may increase typing errors. Based on this speculation, we hypothesized that the performance of 4T in VR can be improved if the virtual cursors can accurately represent the thumb contact points.

We first developed a correction method to estimate the contact points from tracker points to verify our hypothesis. Because 4T on a smartphone is an everyday task, users may have developed a mental model for perceiving contact points from proprioception and muscle memory for hitting correct targets with contact points. Assuming that the gaps between tracker points and contact points are consistent for each key, we developed a correction method to estimate the contact points from the tracker points. We used simple regression to implement the correction method because the correction method should not require large-scale data and a long execution time for a real-time experiment. We then conducted an experiment to show the effect of contact point feedback on the performance of 4T in VR compared to tracker point feedback. The experiment results showed that contact point feedback had a

positive effect on typing performance; participants achieved typing speeds over 25 WPM after a short period of practice.

The improved typing speed owing to contact point feedback may be a small practical contribution. A more significant contribution of the current study is that we have shown that the gap between contact points and tracker points is one of the factors that can explain the gap between the efficiencies of 4T in VR and real 4T – design knowledge for improving 4T in future VR systems.

2 TWO-THUMB TOUCH TYPING IN VIRTUAL REALITY

Many VR systems use handheld controllers to support input tasks in VR, one of which is text entry. To identify efficient and preferable text entry methods in VR, Speicher et al. [18] compared six representative selection-based text entry methods. In their study, four of them (Controller Pointing, Controller Tapping, Discrete Cursor, and Continuous Cursor) utilized handheld controllers. The Continuous Cursor method, commonly supported in VR [4, 5, 9, 10, 18], allows users to set the cursor on a virtual keyboard using the thumb's position over or on a touchpad and select a key by touching or pressing a touchpad. In the Continuous Cursor method, users hold a controller in each hand and type using two thumbs. Therefore, the Continuous Cursor method may be regarded as an example of 4T in VR.

Bai et al. [2] proposed a tracking system that enables touch operations using a smartphone in virtual reality. However, the 4T performance using the system was approximately 10 WPM in a word-typing task, which is much lower than that of a real 4T. To reduce the performance gap between 4T in VR and real 4T, Son et al. [15] investigated the effects of two design factors, hover feedback and grip stability, on the performance of 4T in VR. The study showed that users could perform 4T more efficiently when the hover feedback of thumbs is given, but the achieved typing speed was only 19.19 WPM after one and half hour experiment. The study also showed that participants could type at around 30 WPM after two hours of practice. This result is promising and suggests that users need a lot of time to adapt to 4T in VR, even though they are already experts in real 4T. Our study started with the question of why users cannot easily transfer typing skills in real 4T to 4T in VR, and how we may be able to further improve 4T in VR to a level comparable with real 4T. We focused on the gap between the virtual cursor positions and the thumb contact points to answer these questions.

3 CONTACT POINT FEEDBACK VERSUS TRACKER POINT FEEDBACK

During two-thumb typing on a smartphone, the touchscreen of the smartphone tracks the approximate contact points of the thumbs. These points on the thumbs are closest to the screen. Contact points are not constant because they change depending on the thumb posture, which is different for different target keys during two-thumb typing on a smartphone. However, text input on a smartphone is an everyday task. Therefore, users may have unconsciously learned where their contact points are when touching keys during two-thumb typing on a smartphone. In other words, users may have

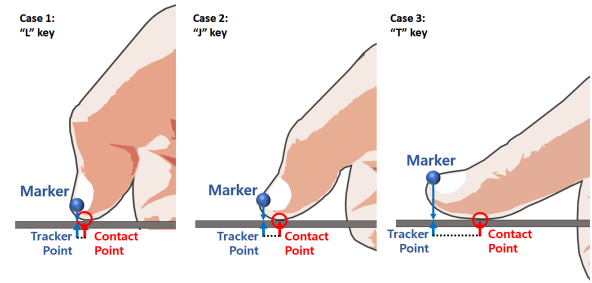


Figure 2: The error between a tracker point and a contact point is affected by the thumb's posture, different for different target keys.

developed a mental model for perceiving contact points from proprioception and muscle memory for hitting the correct target keys with contact points.

However, 4T in VR does not provide the users' real thumbs directly and usually provides virtual cursors visualizing the thumb positions, which are the positions of the reflective markers on the thumbs (tracker points). Tracker points may not correspond to contact points as shown in Figure 1, but users may regard them as contact points. Tracker points may vary from person to person and change depending on where the markers are attached. Therefore, the visual feedback by the virtual cursors may conflict with the contact points that users expect from proprioception and may induce typing errors. Typing errors may frustrate users and disturb typing skills developed in their real 4T experiences. In addition, the direction and degree of errors between the virtual cursors and contact points in the user's mind may not be uniform over the keyboard because the thumb posture changes. Figure 2 illustrates how the errors may differ for different thumb postures. Owing to the irregularity of errors over the keyboard, users may have to pay more attention to the virtual cursors to avoid errors, which may hamper the transfer of their real 4T skills to 4T in VR.

We hypothesized that the virtual cursors accurately representing the thumb contact points may resolve the conflict between the virtual cursor movements and the thumb movements perceived by proprioception. Suppose users understand that virtual cursors correspond to the thumb's contact points. In that case, they may be better able to transfer their real 4T skills to 4T in VR better and achieve better performance with 4T in VR.

4 OBTAINING CONTACT POINTS FROM TRACKER POINTS

Assuming that users use a consistent thumb posture for each key, we decided to design regression functions from a tracker point (x_t, y_t, z_t) to a contact point (x_c, y_c, z_c) by fitting along each axis as we explain below.

The data consisting of (tracker point, contact point) pairs for both thumbs were collected with real 4T. The regression function fitting the data is expected to provide the same contact points of the thumbs in real 4T and bring users' proprioception of the thumbs in real 4T to 4T in VR. When users touched a key during the real 4T, the positions of the reflective markers from the tracker

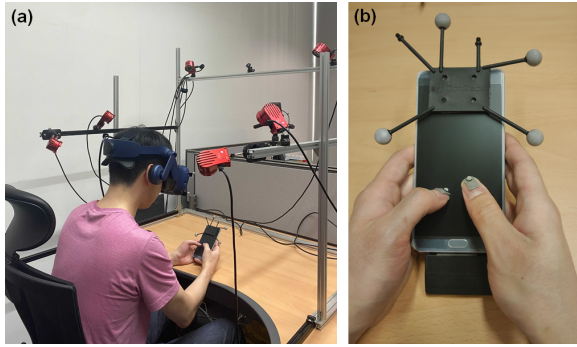


Figure 3: The experimental setup. (a) thumbs tracking system and (b) reflective markers attached to thumbs and a smartphone

and the thumb's contact point positions from the touchscreen were collected. If users made a typing error and corrected it, the data were not included in the fitting dataset. To obtain a smooth regression function, the data gathered in the boundary regions of the keyboard were duplicated to fill the extended space surrounding the keyboard. In addition, because it was impossible to acquire the contact points in the hovering space over the keyboard, data collected on the keyboard were duplicated along the z-axis to fill the space over the keyboard. To obtain a regression function from the collected data, we used the Levenberg–Marquardt algorithm. We utilized the Accord.Net framework [17], which provides the Levenberg–Marquardt algorithm in the C# environment. A cubic polynomial function of the three tracker point variables, x_t , y_t , and z_t , was obtained for each of the contact point variables, x_c , y_c , and z_c , of the left and right thumbs. Therefore, six curve-fitting functions in total were obtained through this process.

The data collection process required a user to type 37 phrases from pangram sentences using real 4T. The mapping from tracker points to contact points may differ for different users. An individual's data were used only for the individual's regression function.

5 EXPERIMENT

The goal of the experiment was to examine the effect of contact point feedback on the performance of 4T in VR. Twelve university students (four females and eight males, average age = 21.9) were recruited. All participants were regular users of a QWERTY keyboard on a smartphone. Each participant was paid approximately 15 USD for participation.

5.1 Apparatus

We utilized Samsung Galaxy Note 5¹ for collecting calibration data and also for the handheld device in the experiment. We used an OptiTrack motion capture system with six OptiTrack Flex 13 cameras² to track thumb movements on and over the smartphone. The cameras were mounted on an aluminum profile structure, as shown in Figure 3a. Reflective markers were attached to the smartphone and on the tips of both thumbs, as shown in Figure 3b. The positions

¹<https://www.samsung.com/global/galaxy/galaxy-note5/>

²<http://optitrack.com/products/flex-13/>

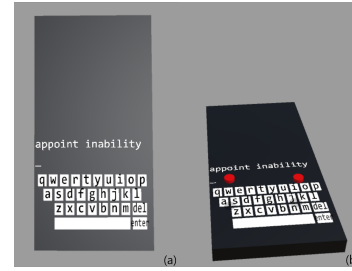


Figure 4: Virtual keyboard utilized in the experiment. (a) a keyboard layout that resembles the basic QWERTY keyboard, (b) the keyboard and virtual cursor from a user's point of view.

of the reflective markers on the thumb nails were chosen carefully so that the resultant tracker points may be as close to the contact points as possible on average. The tracking system was also used in the calibration data collection process using real 4T.

We used a Vive Pro Eye head-mounted display (HMD)³ for the VR environment. The HMD supports a 2880×1600 resolution at a 90 Hz refresh rate and 110° field-of-view. The HMD's position was tracked using the OptiTrack motion capture system.

Figure 4 shows a screenshot of the virtual keyboard we implemented to support the 4T in VR. The red disks represent the positions of the thumbs over the keyboard. When the thumb selected a key by touching the touchscreen, the key below the cursor turned red. The keyboard layout is based on the basic QWERTY keyboard on a smartphone.

5.2 Design

The experiment was a within-subjects design, where the independent variable was *Feedback*, which was Contact Points or Tracker Points. In the Contact Points condition, the positions of the virtual cursors were determined by the contact points. In the Tracker Points condition, the positions of the virtual cursors were determined by the tracker points. The order of the two conditions was counter-balanced across the participants. We used two standard metrics for text entry research, as suggested by Soukoreff and MacKenzie [16]: words per minute (WPM) and corrected error rate (CER).

5.3 Procedure

Participants were asked to attach reflective markers on their thumbs and sit in front of a desk. Before the experiment, the participants were asked to conduct the data collection process following the experimenter's instructions in a real 4T environment to find a personalized regression function. The participants were then given a short introduction to 4T in VR. We did not explain exactly what the virtual cursor represents to prevent participants from preconceiving how they should move their thumbs for typing. Participants completed two sessions, one for each of the two feedback conditions.

Since practicing typing with tracker points feedback may affect their typing skills with contact points feedback and vice versa,

³<https://www.vive.com/us/product/vive-pro-eye/overview/>

		Tracker Points	Contact Points
WPM	Mean	22.21	25.67
	SD	3.58	5.51
CER (%)	Mean	12.98	10.03
	SD	6.15	5.49

Table 1: The means and standard deviations of the two performance metrics in the experiment.

their learning in each condition had to be reset or reduced. For this purpose, participants were asked to conduct three blocks of text entry with a real 4T on a smartphone before each session. After three blocks of real 4T text entries, they were allowed to practice 4T in VR with the feedback condition corresponding to the upcoming session for up to 5 min. They repeated six blocks in each session, and five sentences were randomly selected from a phrase set in each block. Each phrase in the set consisted of two words. We made this decision because more words required a smaller font, which was difficult to recognize in the HMD environment. The words for the phrases were chosen from Kaufman [8], considering the letter frequency in the English language. They were instructed to input as quickly and accurately as possible. They were then forced to break for 1 min between the blocks. After finishing all sessions, they answered a questionnaire consisting of an open-ended question on — whether they perceived a difference between the two sessions. The experiment took approximately one hour per participant.

5.4 Result

Figure 5 shows the root mean square error (RMSE) of the contact point estimation using the regression function of each participant for both thumbs. The average RMSEs of x- and y-position of the left thumb are 0.35 mm (SD: 0.10) and 0.31 mm (SD: 0.12), respectively. The average RMSEs of x- and y-position of the right thumb are 0.30 mm (SD: 0.06) and 0.42 mm (SD: 0.07), respectively. The average RMSEs of z-position of the left and right thumbs are 0.44 mm (SD: 0.16) and 0.44 mm (SD: 0.27), respectively.

Table 1 lists the average WPMs for the two feedback conditions for the six blocks. A paired samples t-test showed that *Feedback* had a significant main effect on WPM ($t = -1.83, df = 22, p < 0.05$). 4T in the Contact Points condition showed a faster text entry speed than 4T in the Tracker Points condition. Therefore, our hypothesis, the contact point feedback will improve 4T in VR, was supported.

The average CERs for the two feedback conditions are listed in Table 1. A paired samples t-test showed that *Feedback* had no significant main effect on CER ($t = 1.24, df = 22, p > 0.1$). The CER result suggests that users did not make fewer typing errors in the Contact Points condition. However, five of the twelve participants answered that they perceived a difference between the two sessions in the questionnaire. All five participants mentioned that they could type faster or more accurately in the Contact Points condition. This was interesting user feedback because they were not told which session corresponded to which feedback condition.

5.5 Discussion

The experiment showed that contact point feedback improved the typing speed of 4T in VR but did not influence the frequency of corrected typing errors. To examine the differences in typing skills made by contact point feedback, we measured the average peak velocity (PV) of thumb movements. We expected that the contact point feedback would allow users to move their thumbs faster because they might understand and be more confident about the location of the contact points. A paired samples t-test showed that *Feedback* had a significant main effect on PV ($t = -2.01, df = 22, p < 0.05$), confirming that users could move their thumbs faster in the Contact Points condition. This may also suggest that users could conduct more ballistic thumb movements, as they could do with real 4T on a smartphone.

Our goal in this study was to show the negative effect of the gap between contact points and tracker points and identify room for improvement for 4T in future VR systems. A simple regression approach to estimate the contact points from the tracker points was sufficiently effective to achieve this goal. However, if our goal was to achieve the best performance by contact points feedback, we had to consider a better approach to estimate the contact points. For example, we can design a method to estimate the contact points based on a perceived input point model from Holz and Baudisch [6, 7].

The regression model in this study was estimated and used for each individual because the mapping from tracker points to contact points was thought to differ for different users. We observed that the locations and distributions of tracker points and contact points varied across participants while developing the regression algorithm. The differences between tracker points and contact points may be affected by a few factors, such as the thickness of the thumbs and typing habits that determines where to touch and which posture to use. Therefore, we concentrated on showing the effect of contact point feedback in this study, rather than developing a generalized estimation method that can consider all the individual differences. Finding a generalized estimation model of contact points should be considered in the future to apply the conclusion of this study to practical VR systems.

The regression approach to estimate contact points was possible because we used an accurate marker-based tracker to track the fingers. Commodity finger tracking systems for VR HMDs, however, do not use a marker-based tracker but uses a vision-based tracker, which cannot provide an accurate and fast estimates of finger positions. Researchers showed that commodity finger tracking systems yield high tracking errors between the finger and the target [13, 14]. With such large tracking errors, the cursors cannot reflect the dynamics of the thumbs, and therefore the discrepancy between the cursor movements and users' proprioception may not be an issue any more. In other words, users need to type relying solely on the cursors in this case, without the chance to utilize the muscle memory that they developed using real 4T. In this respect, the practical value of the current study depends on the availability of an accurate finger tracking technology in VR. Considering the rapid growth of VR technology, we expect finger tracking of VR HMDs will improve soon to support our approach. Alternatively, a separate, specialized technology for accurate finger tracking may be

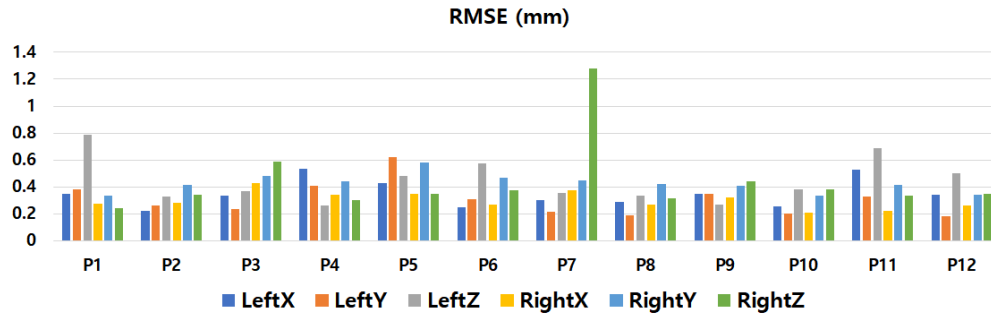


Figure 5: The root mean square error (RMSE) of contact point estimation by the regression function.

employed, such as the deep-learning-based finger tracking method by Matulic et al. [11].

In this study, the positive effect of contact points feedback was shown in the case of letter-wise text entry. Contact point feedback may not create a statistically significant difference when auto-correction and/or word suggestion feature are used, which is common in modern keyboards. Further study is needed to determine whether contact point feedback is essential for a keyboard with auto-correction and/or word suggestion feature. However, we believe that the positive effect for the letter-wise input case alone has practical importance, considering that it is being discussed whether auto-correction and word suggestion have clear benefits over letter-wise input [3, 12].

While other approaches [4, 5, 18] have not achieved a typing speed of over 25 WPM using a touchpad controller in VR, 4T in VR in this study showed a text entry speed of 25.67 WPM in a short period of time. Considering the results of Son et al. [15] and ours, we expect that 4T can be one of the best text entry methods in VR in terms of typing performance. On the other hand, we could still observe a large gap between 4T in VR and real 4T in this study; in the short real 4T sessions before two 4T in VR sessions, participants showed an average typing speed of 38.13 WPM. Further investigation of additional factors to explain the gap between 4T in VR and real 4T is needed to improve 4T in future VR systems.

6 CONCLUSION

We aimed to support a better transfer of real 4T skills to 4T skills in VR by providing contact point feedback. We developed a simple regression method to estimate contact points from tracker points to verify this possibility. We then investigated the effect of contact point feedback on the performance of 4T in VR. The experimental results showed that contact point feedback positively impacted the typing speed of 4T in VR. In particular, participants could achieve a typing speed of over 25 WPM after a short practice period. This is an encouraging result; however, there is still a significant performance gap between 4T in VR and real 4T. We believe that 4T in VR has much room for further improvement and may achieve the efficiency of real 4T in the future.

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